Artificial Intelligence-II – Semester 5

Practical 1: Write a program to implement Single Player Game (Using Heuristic Function)

Why is it a heuristic?

A **heuristic function** is a function that estimates how close the current state is to the goal state.

In your game:

- **Current State** = Player's current guess
- **Goal State** = Correct number
- **Heuristic Function**

h(n)=∣Target−Guess∣h(n)=|Target-Guess|h(n)=∣Target−Guess∣

This value tells us the estimated distance from the goal.

Example (Target = 87):

Guess	Heuristic h(n)
20	67
50	37
80	7
85	2
87	0

Notice:

- As the player gets closer, the heuristic value decreases.
- When the heuristic becomes 0, the goal is reached.

```
import random
target = random.randint(1, 100)
print("==================================")
print("  NUMBER GUESSING GAME (AI)")
print("==================================")
print("Guess a number between 1 and 100")

attempts = 0

while True:
    guess = int(input("\nEnter your guess: "))
    attempts += 1

    # Heuristic Function
    difference = abs(target - guess)

    if guess == target:
        print(f"\n🎉 Congratulations!")
        print(f"You guessed the correct number: {target}")
        print(f"Total Attempts: {attempts}")
        break

    print(f"Heuristic Value (Difference from Target): {difference}")

    if guess < target:
        print("Hint: Try a Higher Number")
```

```
else:
    print("Hint: Try a Lower Number")
```